

5(a)	Apply list rule for 4 responses $C^P C^P NN$ $C^P C^Y NN$ $C^P C^P Nn$ $C^P C^Y Nn ; ;$	2	two or three correct = 1 mark four correct = 2 marks
5(b)(i)	<i>genotype:</i> $C^B C^Y Nn ;$ <i>phenotype:</i> brown without stripes ;	2	If both correct answers given but on wrong lines give 1 mark ECF for a phenotype that matches a wrong genotype
5(b)(ii)	<i>any four from:</i> 1 independent / random, assortment / orientation (of chromosomes / alleles / genes) ; 2 in metaphase I ; 3 <i>ref. to</i> genes, on separate chromosomes / not linked OR no linkage ; 4 gametes / (daughter) cells / products of meiosis, are, haploid / n ; 5 $2^2 / 4$, combinations, in dihybrid cross / with two genes OR (get) $C^B N \ C^B n \ C^Y N \ C^Y n ;$	4	IGNORE crossing-over 1 ALLOW each pair of, chromosomes / alleles, lines up / separates / segregates, independently of other pairs 2 ALLOW anaphase I if segregation described 2 DO NOT ALLOW metaphase / anaphase, II 3 CON <i>ref. to</i> breaking of linkage groups 4 ALLOW get one of each pair of, chromosomes / alleles for 'haploid / n' 5 ALLOW digenic for 'with two genes'

Question	Answer	Marks	Guidance
5(b)(iii)	any three from: 1 (autosomal) linkage OR genes were, linked / on same chromosome ; 2 C^B n, inherited together / linked / on same chromosome, and C^Y N, inherited together / linked / on same chromosome ; 3 (offspring showed) parental phenotypes ; 4 no, recombinants / crossing-over ; 5 as, genes / loci, tightly linked / close together ;	3	1 CON sex linkage 2 ALLOW words e.g. <u>alleles</u> for (brown + stripes) and (yellow + no stripes), inherited together / linked / on same chromosome 3 phenotypes were same as original parents

Question	Answer	Marks	Guidance
5(c)	any four from: 1 in woodland, non-yellow / brown / pink / unstriped, snails, are not seen / are camouflaged / are hidden / are not eaten / survive OR in rough grass, yellow / striped, snails, are not seen / are camouflaged / are hidden / are not eaten / survive ; 2 in woodland, non-yellow / brown / pink / unstriped, snails have a <u>selective advantage</u> / are <u>selected for</u> OR in rough grass, yellow / striped, snails have a <u>selective advantage</u> / are <u>selected for</u> ; 3 in woodland, non-yellow / brown / pink / unstriped, snails, reproduce / pass on alleles OR in rough grass, yellow / striped, snails, reproduce / pass on alleles ; 4 in woodland, non-yellow / brown / pink / unstriped, snails, increase in frequency OR in rough grass, yellow / striped, snails, increase in frequency ; 5 in woodland, $C^B / C^P / N$, are selected for / increase (in frequency) OR in rough grass, C^Y / n, are selected for / increase (in frequency) ;	4	ALLOW ORA throughout ECF after first CON to max 3 e.g. in rough grass brown snails selected for, survive and reproduce = CON mp2 then gets mp 1 and mp 3. 2 ORA non-yellow / brown / pink / unstriped, snails <u>selected against</u> in rough grass OR yellow / striped, snails <u>selected against</u> in woodland 2 ALLOW selection pressure favours for 'selected for' 2 ALLOW selection pressure is against for 'selected against' 3 ALLOW pass on genes (to offspring / next generation) for 'reproduce' 4 ORA in rough grass, non-yellow / brown / pink / unstriped, snails, decrease in frequency OR in woodland, yellow / striped, snails, decrease in frequency 5 ALLOW words e.g. <u>allele</u> for pink (shell) for 'C^P'